



INTERNATIONAL MATHEMATICAL OLYMPIAD

DA-MATH 220 FINAL PROJECT, SPRING 2025

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BACKGROUND AND INTRODUCTION

Goal:

Explore what factors influence a country's performance at the International Mathematical Olympiad (IMO).

Why This Topic?

- We want to understand how team composition and national education scores relate to success.

Countries Focused On:

- The Netherlands **NL**
- United States **US**
- People's Republic of China **CN**



WHY FOCUS ON THE FACTORS PISA SCORES AND FEMALE PARTICIPATION?

- PISA (Programme for International Student Assessment) measures national levels in math, reading, and science among 15-year-olds.
 - IMO participants are typically high school students
- The IMO has historically had low female representation
 - Investigating how gender diversity affects IMO performance to discover how more female participants might impact overall scores.



OUR QUESTIONS

Main Question:

What impacts a country's performance at the IMO?

Sub-Questions:

How does team composition (especially gender diversity) affect performance?

How are national PISA scores related to IMO success?



DATA SOURCES

IMO Data:

- Source: [International Math Olympiad \(1959-2024\)](#)
- Over 3,500 Observations since 1959

PISA Data:

- Source: [World Population Review \(PISA 2022 Scores\)](#)
- 81 Observations for 81 Participating Countries



INTERNATIONAL OLYMPIAD VARIABLES

Code	Description
team_size_all	Total people on one competing team
team_size_male / team_size_female	How many males / females on one team
team_IMO_Score	The total amount of points added up from all 6 problems given in the competition
awards_gold/bronze/silver	Amount of gold, bronze, and silver medals won per team in the competition
Country	Name of the country



PISA VARIABLES

Code	Description
OverallPisaScore2022	Total PISA score in 2022
PISAScoresMathScore2022	Points received in the math category
PISAScoresScienceScore2022	Points received in the science category
PISAScoresReadingScore2022	Points received in the reading category

STATISTICAL METHODS

Linear Regression:

- What impacts a country's performance at the International Mathematical Olympiad

Likelihood Methods:

- Predict future country performance based on historical patterns

T-tests:

- How does team composition (especially gender diversity) affect performance?

Correlation:

- How are national PISA scores related to IMO success?

Visualization



PRELIMINARY EXPECTATIONS

Countries with higher PISA math scores likely perform better at the IMO.

A light red arrow pointing downwards from the first box to the second box.

Gender-balanced teams may show different performance trends.

A light yellow arrow pointing downwards from the second box to the third box.

China's dominance expected to be reflected both in PISA and IMO results.

HOW DOES TEAM COMPOSITION (ESPECIALLY GENDER DIVERSITY) AFFECT PERFORMANCE?

Team IMO Score (Team_IMO_Score) - *The total amount of points added up from all 6 problems given in the competition*

Female Contestants (team_size_female) - *The total amount of female contestants on each team that competes*

Country	Count of Teams All Time With 0 Female Contestants	Count of Teams All Time With At least 1 Female Contestant
China	30	9
USA	42	6
Netherlands	38	15





GENDER DIVERSITY HYPOTHESIS TEST: CHINA

Null: H_0 - There is no difference in means in Team IMO Score when there are female contestants representing

Alternate: H_1 - There is a difference in means in Team IMO Score when there are female contestants representing

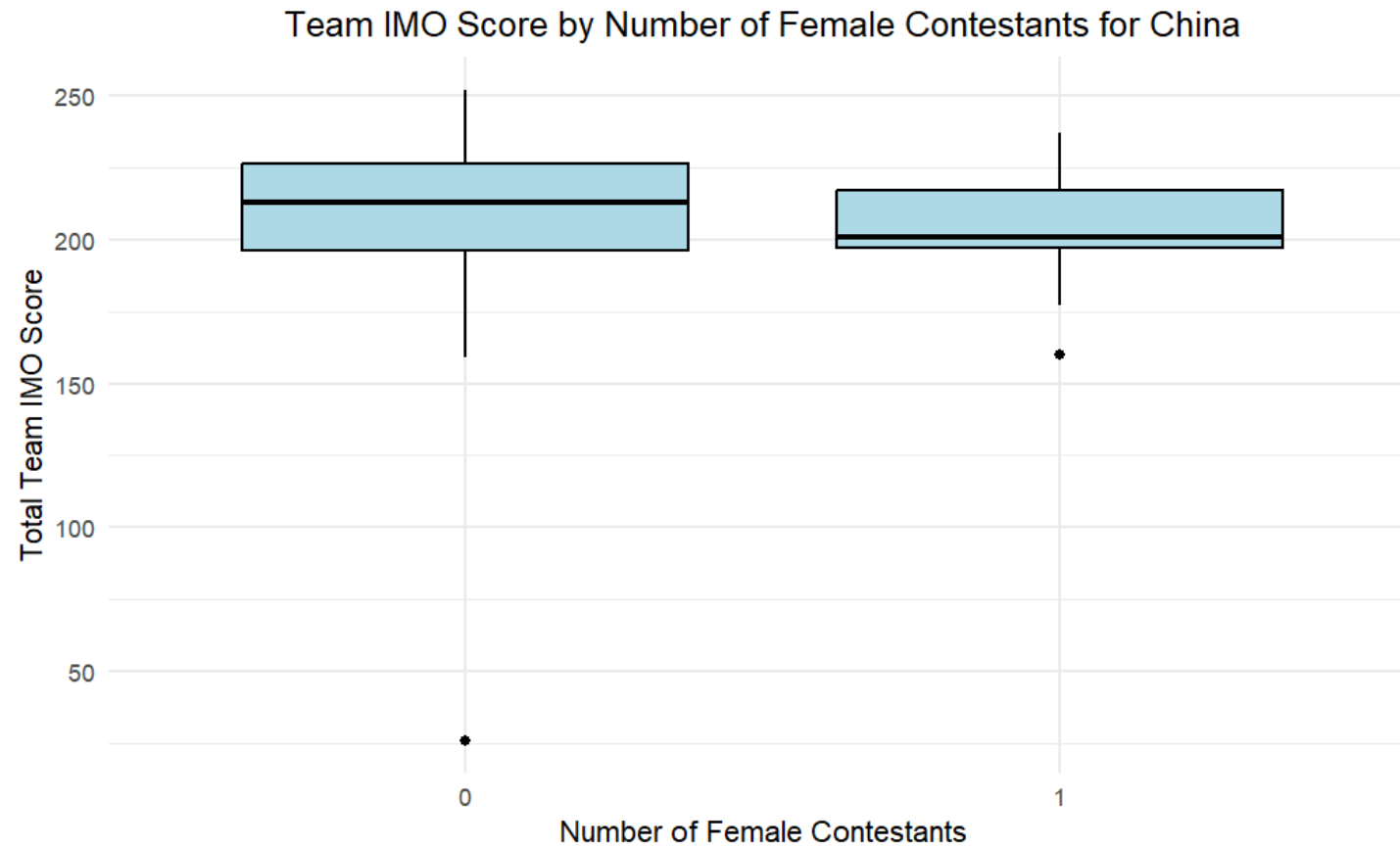
Measure	Result
Mean (Female Present)	204.44
Mean (No Female)	205.53
95% Confidence Interval	(-24.27, 22.09)
Degrees of Freedom	21.06
p - value	.923



Not Statistically Significant!



GENDER DIVERSITY BOXPLOT: CHINA





GENDER DIVERSITY HYPOTHESIS TEST: USA

Null: H_0 - There is no difference in means in Team IMO Score when there are female contestants representing

Alternate: H_1 - There is a difference in means in Team IMO Score when there are female contestants representing

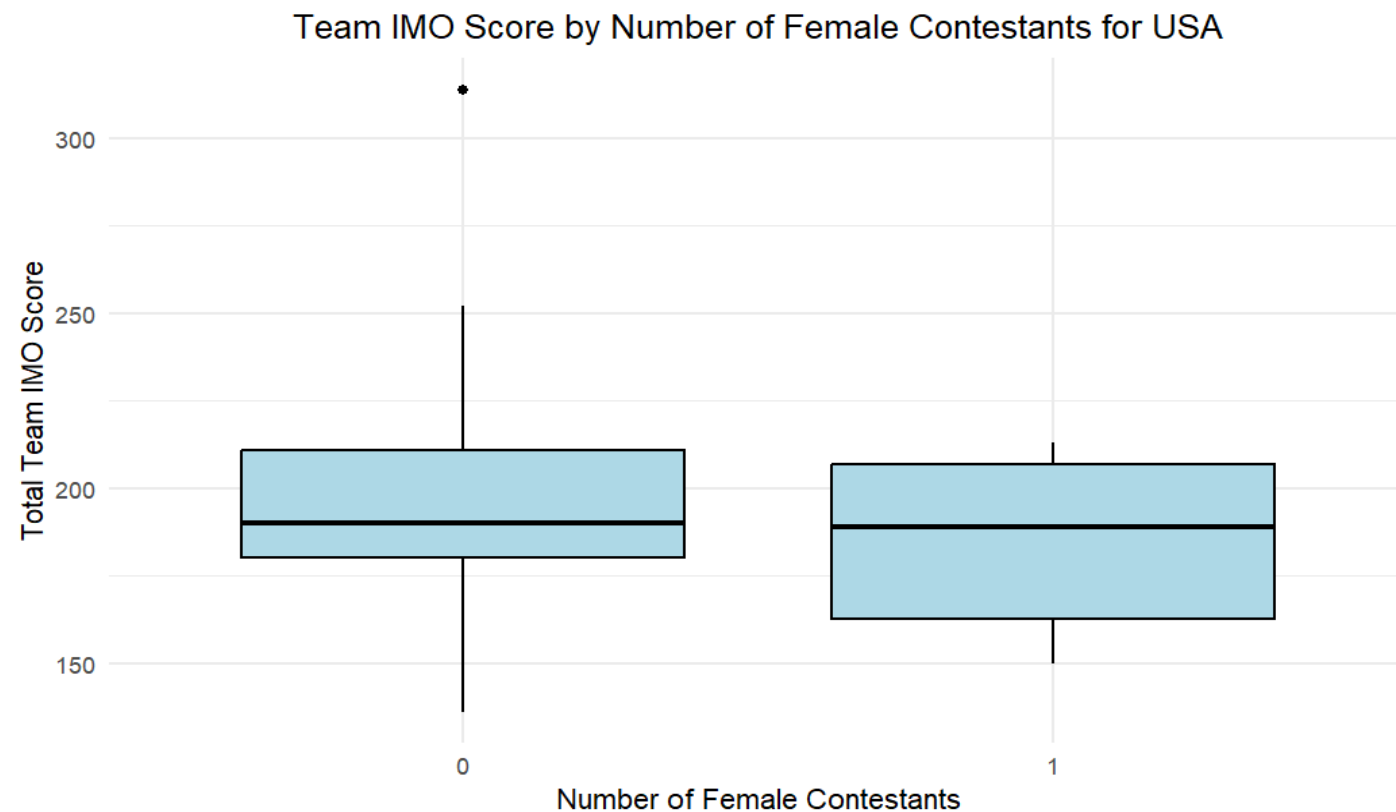
Measure	Result
Mean (Female Present)	184.67
Mean (No Female)	195.29
95% Confidence Interval	(-39.20, 17.97)
Degrees of Freedom	7.17
p - value	.410



Not Statistically Significant!



GENDER DIVERSITY BOXPLOT: USA



GENDER DIVERSITY HYPOTHESIS TEST: NETHERLANDS

Null: H_0 - There is no difference in means in Team IMO Score when there are female contestants representing

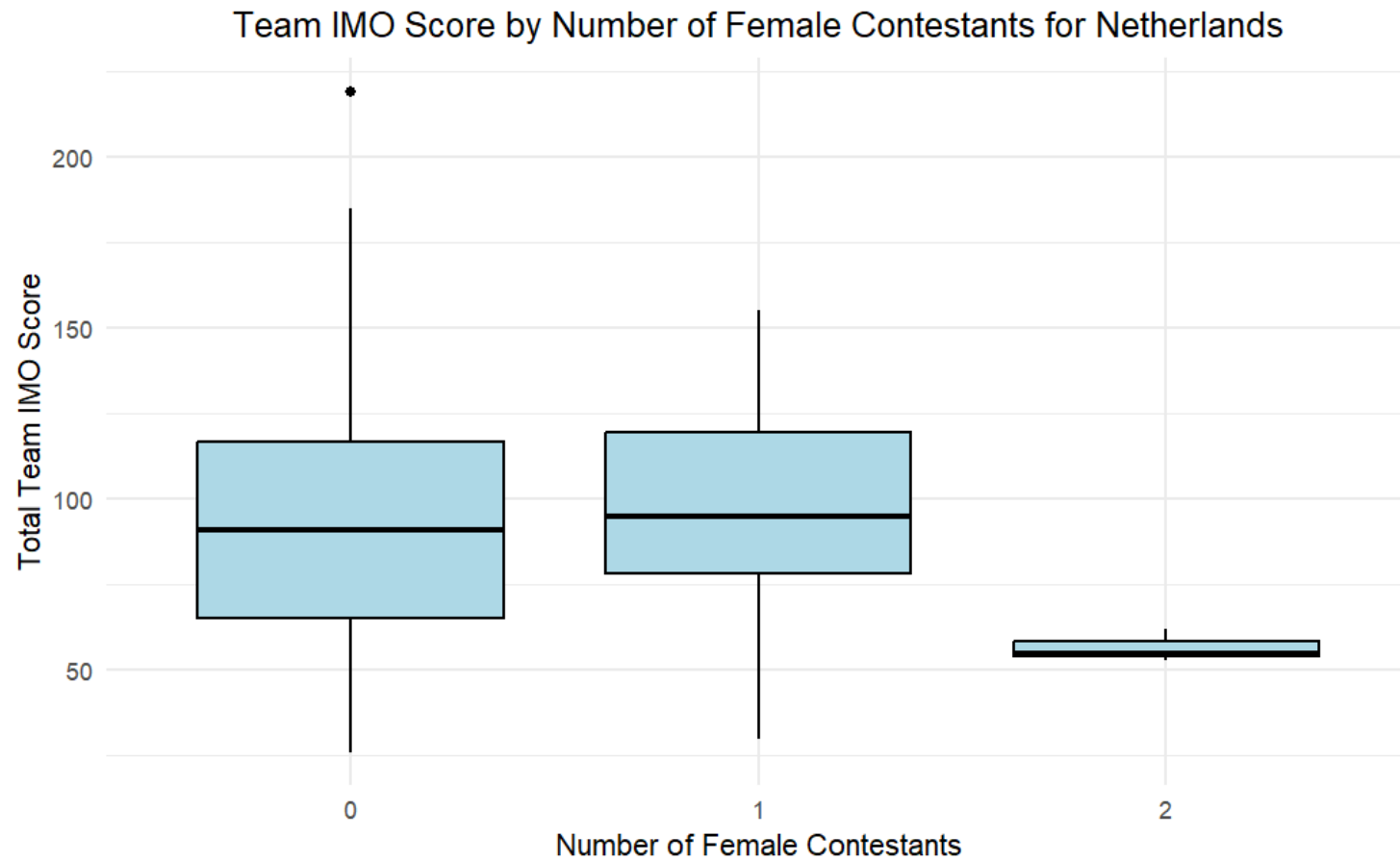
Alternate: H_1 - There is a difference in means in Team IMO Score when there are female contestants representing

Measure	Results
Mean (Females Present)	90.60
Mean (No Females)	95.32
95% Confidence Interval	(-27.73, 18.30)
Degrees of Freedom	30.13
p - value	.679



Not Statistically Significant!

GENDER DIVERSITY BOXPLOT: NETHERLANDS



GENDER DIVERSITY CONCLUSIONS

- Majority of teams all time have not included at least one female contestant
- China had the highest average Team IMO Score like we expected in our preliminary
- From the tests ran we can potentially conclude and predict that having at least one female contestant will not influence the Total IMO Score for the team in any way (All p-values were greater than significance level of .05)
- No official conclusions can be made since we are only choosing 3 countries as our sample size (US, CN, NL) and there can be potential bias from these selections.



RELATIONSHIP BETWEEN INTERNATIONAL MATH
COMPETITION PERFORMANCE AND PISA SCORES
(2 0 2 2)

- PISA 2022 Results: International standardized test scores for students (Math, Science, Reading, Overall scores).
- IMO 2022 Results: Scores from the International Mathematical Olympiad.

flagCode	OverallPisaScore2022	PISAScoresMathScore2022	PISAScoresScienceScore2022	PISAScoresReadingScore2022
CN	1605	552	543	510
US	1468	465	499	504
NL	1440	493	488	459

Team_IMO_Score
252
207
145



CORRELATION

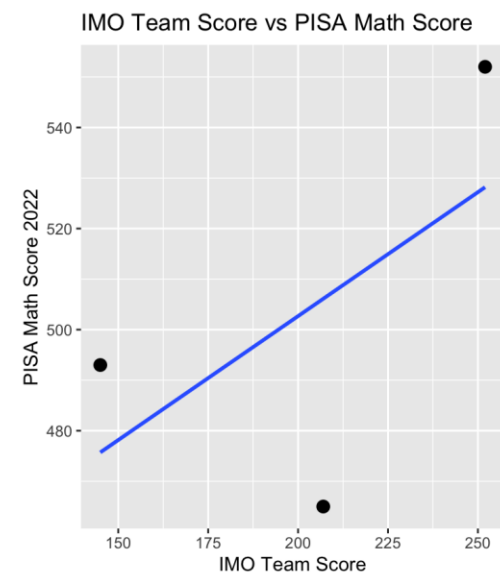
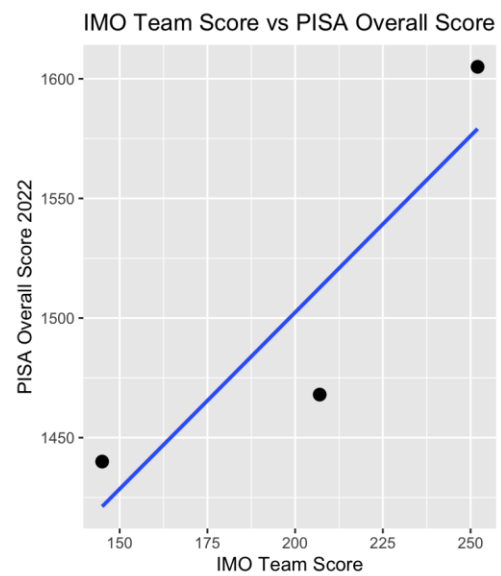
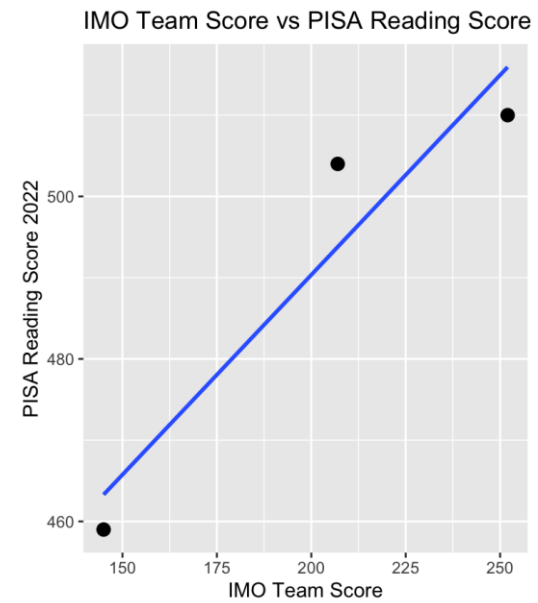
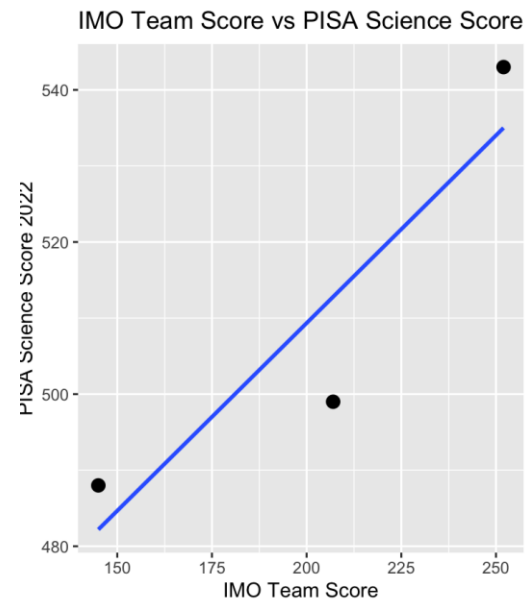
PISA Domain	Correlation with IMO Team Score
Overall Score	0.8979
Math Score	0.5932
Science Score	0.9111
Reading Score	0.9479



INTERPRETATION OF RESULTS

- Strong relationships are observed with the PISA Reading Score (0.948) and the PISA Science Score 2022 (0.911)
- The correlation between the Team IMO Score and the Overall PISA Score 2022 is slightly lower (0.898) but still strong.
- PISA Math Score 2022 shows a moderate correlation (0.593) with the Team IMO Score





REFLECTION ON IMO-PISA

- With only three data points per domain, correlation coefficients are fragile
 - A sample size of three countries is too small for any real statistical power
 - **Improvement: Expand the sample to include 20-30 countries**
- Current data only consists of one year of IMO and PISA performance
 - Analyzing multiple years might help with determining if the correlations are consistent over time



CONCLUSION/KEY TAKE-AWAYS

- **Gender diversity** in IMO teams (US, China, NL) **did not significantly impact** team scores – all p-values > 0.05
- **Strong positive correlations** were found between 2022 IMO performance and PISA scores, especially:
 - Reading (0.948)
 - Science (0.911)
 - Overall and Math scores also showed high to moderate correlation.



THANK YOU!

